Packet Tracer – Static IP and Subnet Assignment

Objectives

Part 1: Build a Simple Network Topology

Part 2: Configure Static IP Addresses

Part 3: Practice Different Subnet Masks

Part 4: Verify Connectivity and Understand Subnetting

Background / Scenario

You are practicing one of the most important networking concepts: **subnetting**. While IP addresses identify devices on a network, subnet masks determine which devices belong to the same local network.

In this lab, you will connect two PCs, manually assign their IP addresses, and experiment with different subnet masks to observe how they affect communication.

This lab focuses on:

- Assigning static IPv4 addresses
- Understanding subnet masks
- Determining network membership
- Testing connectivity using ICMP (ping)

You are building the foundation necessary for future routing, VLAN, and CCNA subnetting labs.

Instructions

Part 1: Connect the Devices

Step 1: Add the PCs

a. Open **Cisco Packet Tracer**.

b. From **End Devices**, drag a **PC** onto the workspace.

c. Drag another **PC** onto the workspace.

d. Rename them:

- PC0 → **PC-A**
- PC1 → **PC-B**

Step 2: Connect the Two PCs

Since this is a direct connection between two similar devices, you will use a **Copper Cross-Over** cable.

a. Click **Connections** (lightning bolt icon).

b. Select **Copper Cross-Over**.

c. Connect **PC-A → FastEthernet0**.

d. Connect **PC-B → FastEthernet0**.

Wait a few seconds until both link lights turn **green**.

Your physical topology is now complete.

Part 2: Configure Static IP Addressing

Step 1: Configure PC-A

a. Click **PC-A**.

b. Select the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.10.10**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

Step 2: Configure PC-B

a. Click **PC-B**.

b. Select the **Desktop** tab.

c. Click **IP Configuration**.

d. Select **Static**.

Enter:

- IP Address: **192.168.10.20**
- Subnet Mask: **255.255.255.0**
- Default Gateway: *(leave blank)*

Close the window.

Why Must the Subnet Masks Match?

The subnet mask tells a device which portion of the IP address represents the network and which portion represents the host.

Since both PCs are configured with:

- Network Address: **192.168.10.0**
- Subnet Mask: **255.255.255.0 (/24)**

they recognize each other as members of the same local network and can communicate directly.

Part 3: Verify Connectivity

Step 1: Ping from PC-A to PC-B

a. Click **PC-A**.

b. Go to the **Desktop** tab.

c. Open **Command Prompt**.

d. Type:

ping 192.168.10.20

Press **Enter**.

Expected Result:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

You should receive **4 successful replies**.

Step 2: Ping from PC-B to PC-A

a. Click **PC-B**.

b. Open **Command Prompt**.

c. Type:

ping 192.168.10.10

Press **Enter**.

You should again receive **4 successful replies**.

Part 4: Explore Different Subnet Masks

Now you will observe how changing the subnet mask affects communication.

Experiment 1: Use a Smaller Subnet

Change the configuration of both PCs to:

Device	**IP Address**	**Subnet Mask**
PC-A	192.168.10.10	255.255.255.192
PC-B	192.168.10.20	255.255.255.192

Test the ping again.

Result: The ping should still succeed because both devices belong to the subnet:

192.168.10.0/26

Experiment 2: Place Devices in Different Subnets

Keep PC-A unchanged:

- IP Address: **192.168.10.10**
- Subnet Mask: **255.255.255.192**

Change PC-B to:

- IP Address: **192.168.10.70**
- Subnet Mask: **255.255.255.192**

Try pinging again.

Expected Result:

Request timed out.

The ping fails because:

- PC-A belongs to network **192.168.10.0/26**
- PC-B belongs to network **192.168.10.64/26**

Without a router, devices on different subnets cannot communicate.

Understanding the Subnetting Impact

Subnet Mask	Prefix	Hosts per Network
255.255.255.0	/24	254
255.255.255.128	/25	126
255.255.255.192	/26	62
255.255.255.224	/27	30

As the subnet mask becomes larger, the number of available host addresses decreases, creating more but smaller networks.

Troubleshooting Tips

If the ping fails:

✓ Verify both PCs are connected using a **Copper Cross-Over** cable.

✓ Ensure the link lights are green.

✓ Confirm both IP addresses are entered correctly.

✓ Check that the subnet masks match the intended network.

✓ Remember that devices on different subnets require a router to communicate.

What You Learned

- How to manually configure static IPv4 addresses.
- The purpose of a subnet mask.
- How subnet masks determine network boundaries.
- Why devices must belong to the same subnet for direct communication.
- How ICMP (ping) can be used to test connectivity.
- That routers are necessary for communication between different networks.

Challenge Extensions (Optional)

Try modifying the lab:

1. Change both PCs to use the subnet mask:

255.255.255.128

   - Can they still communicate?

2. Configure:
   - PC-A: 192.168.10.100/25
   - PC-B: 192.168.10.200/25
   - Do they belong to the same network?
3. Add a **Router** between the two PCs and configure separate subnets.
4. Calculate the following manually before testing in Packet Tracer:

- Network Address
- First Host
- Last Host
- Broadcast Address

for the subnet:

192.168.10.64/26